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High-throughput and Automated Anion Transport Assays

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Abstract

Ion transport across biological lipid membranes is crucial for both health and disease, prompting investigation into synthetic ion transporters as potential therapeutics. However, state of the art methods for assessing transporter activity are often time-consuming and resource-intensive, hindering the screening of large compound libraries. In this work we have developed an automated high-throughput assay capable of determining 24 EC50 values in a single day. This assay, adapted from the established lucigenin Cl-/NO3- antiport method, has been optimised for multi-well plates with fully automated liquid handling. Notably, our protocol allows for the use of DMSO as a delivery solvent for target molecules, enhancing its versatility. Validation with six standard ion transporters demonstrated a strong correlation between the EC50 values obtained through our assay and previously reported values, confirming its efficacy and potential for broader applications in the field of ion transport.

Keywords

ion transport, automation, high throughput, fluorescence, vesicle

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High-throughput and Automated Anion Transport Assays

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Abstract

Ion transport across biological lipid membranes is crucial for both health and disease, prompting investigation into synthetic ion transporters as potential therapeutics. However, state of the art methods for assessing transporter activity are often time-consuming and resource-intensive, hindering the screening of large compound libraries. In this work we have developed an automated high-throughput assay capable of determining 24 EC_{50} values in a single day. This assay, adapted from the established lucigenin Cl^-/NO_3^- antiport method, has been optimised for multi-well plates with fully automated liquid handling. Notably, our protocol allows for the use of DMSO as a delivery solvent for target molecules, enhancing its versatility. Validation with six standard ion transporters demonstrated a strong correlation between the EC_{50} values obtained through our assay and previously reported values, confirming its efficacy and potential for broader applications in the field of ion transport.

Introduction

Synthetic ion transporters play a crucial role in the translocation of ions across the phospholipid bilayer, and their biological applications have been investigated for the treatment of channelopathies, cancer, and microbial infections.[1–21] Despite their therapeutic potential, predicting the ion-transport efficiency of novel receptors remains challenging. Previous efforts to address this issue have included quantitative structure-activity relationship (QSAR) studies involving extensive transporter libraries aimed at developing predictive models; however, these approaches are often time-consuming and costly under current methodologies.[22] Notably, the Hill analysis—commonly employed to determine the EC_{50} as a quantitative measure of transport efficacy—requires significant time investment, typically allowing only one transporter to be assessed per researcher per day. Furthermore, repeat experiments are infrequently conducted, and the reproducibility of results is often unexplored. The associated time and financial constraints can be particularly limiting for smaller research groups or those with restricted funding, potentially hindering commercialisation efforts.

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In contrast, high-throughput experimentation has transformed the pharmaceutical sector,[23] yet to our knowledge, no high-throughput methods currently exist for assessing anion transport activity. Such methodologies hold great promise for alleviating the experimental bottlenecks associated with traditional assays, enhancing time efficiency, and enabling the evaluation of more extensive compound libraries, which can lead to quicker hit identification and more comprehensive structure-activity relationship studies. There is an increasing trend towards the integration of automated workflows as cost-effective and productivity-enhancing strategies within medicinal chemistry, fuelled by advancements in automated synthesis, purification, and biological assessment.[24]

In this work, we present the design of high-throughput anion antiport assays that significantly enhance the throughput of ion transport workflows. We have adapted the established lucigenin assay[25] into high-throughput formats utilising commonly accessible fluorescence plate reader technology. Initially, we developed a manual preparation method for a 96-well plate format using multi-channel pipettes, followed by creating semi-automated and automated workflows for 384-well plates employing the Opentrons OT-2 liquid handling platform. Our assays have been benchmarked against literature compounds previously evaluated via ion-selective electrode assays, demonstrating a strong correlation (R^2 = 0.99 - 0.89). This publication aims to provide detailed methodologies and the necessary files for plate-reader setup and data analysis, thereby making these methods accessible to other researchers. We anticipate that these techniques will facilitate the rapid evaluation of extensive compound libraries, enabling future studies to explore a wider chemical space and align more closely with the methodologies favoured by the pharmaceutical industry.

Results and Discussion

Ion transport is typically measured using ion-selective electrode assays or fluorescence assays, and we determined that the latter would be more suitable for adaptation to a high-throughput format. The lucigenin assay, a well-established method in the field, quantifies direct chloride transport across synthetic vesicle membranes, which are usually composed of 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine (POPC) or POPC/cholesterol mixtures.[25] These vesicles contain $NaNO_3$ and lucigenin (*N,N'*-dimethyl-9,9'-biacridiniumdinitrate), a fluorescent sensor that is quenched by chloride ions in a concentration-dependent manner, as described by the Stern-Volmer equation (**equation S1**). The assay begins with the addition of NaCl to the external buffer of the vesicles, allowing the anion transporter to facilitate passive chloride influx via a Cl^-/NO_3^- antiport mechanism, leading to fluorescence quenching. The fluorescence intensity can be calibrated against reference measurements; typically, 0% influx corresponds to the initial fluorescence shortly after the NaCl spike, while 100% influx is represented by the fluorescence reading obtained after adding detergent or a highly effective transporter.[26,27] Various data processing methods can then be employed to quantify transport activity, such as calculating the EC_{50} or the specific initial rate.[25]

The EC_{50} is commonly reported as a measure of transport activity. It reflects the theoretical concentration of an anionophore required to achieve 50% ion influx or efflux by the end of the assay. This value is determined from a Hill plot, which is created by plotting the final chloride influx against transporter concentration. Traditionally, these experiments are performed individually in a cuvette or single vial, typically testing 6-8 different transporter concentrations in triplicate. As a result, this standard protocol usually allows for the assessment of only one compound per day due to time limitations.

To establish a high-throughput (HT) Hill plot assay in a 96-well plate format, we made several adaptations to existing protocols. While the traditional lucigenin assay continuously records fluorescence, the Hill plot only requires an end-point reading; thus, we focused on optimising the accuracy of this single measurement, which was taken five minutes after NaCl addition. Another key objective was to develop an assay compatible with DMSO as a delivery solvent for the anion transporter. DMSO is widely used in other ion transport assays due to its ability to dissolve a broad range of poorly soluble transporters, but it is rarely incorporated into the lucigenin assay because it is known to quench lucigenin fluorescence. Our investigations revealed that the assay could tolerate small amounts of DMSO (as long as the concentration was consistent across wells), with fluorescence partially quenched at 1% v/v. Notably, an intravesicular chloride concentration of 100 mM resulted in a six-fold fluorescence quenching response, even in the presence of 1% DMSO, and the Stern-Volmer plot remained linear

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from 0-100 mM [Cl^-] under assay conditions (**Supplementary Figure S28**). Therefore, by carefully controlling the DMSO concentration and the ionic gradients, we successfully utilised DMSO as a delivery solvent for our transporters.

The optimised assay protocol is outlined in **Figure 1** and involves the following steps: 1) Vesicles containing encapsulated lucigenin were added to each well of a 96-well plate using a multi-channel pipette (150 L per well, 2 mM POPC, internal buffer = 100 µM lucigenin in 200 mM $NaNO_3$ with 5 mM sodium phosphate buffer, pH 7.2. External buffer = 200 mM $NaNO_3$ with 5 mM sodium phosphate buffer, pH 7.2.). Then a solution of transporter in DMSO was added to the lipid, followed by mixing using a multi-channel pipette (3 µL of 100X stock solution). 2) Analysis was performed on a plate reader equipped with an autoinjector, which facilitated the addition of a sodium chloride spike to each well (150 µL per well, 200 mM NaCl, with 5 mM sodium phosphate buffer, pH 7.2). 3) The fluorescence emission intensity (excitation = 369 nm, emission = 503 nm) of each well was measured 5 minutes after the addition of the sodium chloride, and this was processed to give the final Hill plots. Data processing included the creation of a Stern-Volmer plot (**Figure S28**), which was used to convert the fluorescence intensity to a chloride concentration. It is important to note that up to 20% lucigenin leakage from the vesicles may occur within two hours,[25] which should not be confused with ion transport. Additional data calibration was performed using vesicles treated with DMSO only as a reference for 0% chloride influx and a highly potent transporter as a reference for 100% chloride influx (**Equation S2**).

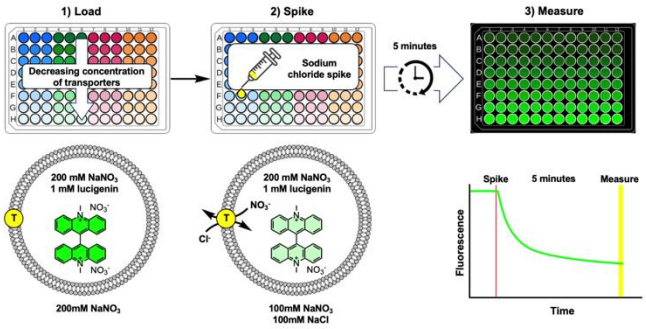


Figure 1. An overview of the steps involved in the high-throughput screening of ion-transporters via the lucigenin assay. This involves adding the transporter solutions to the vesicles and mixing, then adding sodium chloride to the external buffer to begin the assay. The end-point measurements were timed at 5 minutes after the injection of the spike.

To validate our assay, we aimed to obtain the EC_{50} values for known compounds and compare them with literature values. A library of six established ion transporters (**1**–**6**, as shown in **Figure 2**) were synthesised as standards. This library includes anion transporters from various structural classes (such as ureas, thioureas, and squaramides) with reported EC_{50} values spanning three orders of magnitude (from 0.005 to 10 mol% with respect to lipid).[22, 28, 31] The chloride-nitrate antiport activity of these transporters was previously assessed using ion-selective electrode assays.

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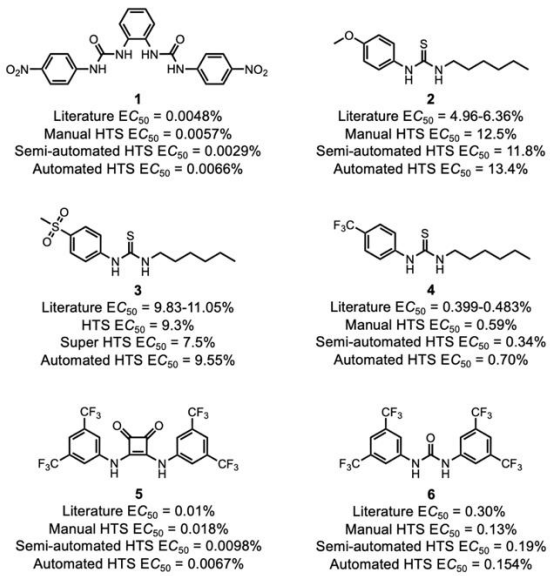


Figure 2.The compound library, their reported E*C*$_{50}$ and the E*C*$_{50}$ found via the high throughput assays

The compound library was initially screened using 96-well plates, and we conducted three Hill plots for each of the compounds **1**–**6** (**Figure 3**). To evaluate the reproducibility and experimental error of our measurements, each Hill plot for a specific compound was performed with a different batch of vesicles on separate plates. The E*C*$_{50}$ values obtained by this method correlated strongly with literature values ($R^2 = 0.99$).

With the initial method established, we focused on automating the most labour-intensive aspect of the assay: the preparation of the multi-well plates. The final composition of each well was standardised, and all volumes were scaled down to one-third to accommodate the smaller capacity of 384-well plates compared to 96-well plates. We developed a liquid dispensing protocol using the Opentrons OT-2 liquid handling robot, with the goal of minimising pipetting errors and enhancing data throughput by utilising 384-well plates. The protocols were crafted using the Opentrons Protocol Designer, and the method files are provided as supplementary material to this article.

The automation process was implemented in stages. Initially, we created a semi-automated protocol that allowed the Opentrons to dispense pre-mixed lipid and transporter solutions into the 384-well plate. Subsequently, we developed a fully automated method where the Opentrons performed all liquid handling steps, including diluting transporter stock solutions in DMSO, mixing the transporter solutions with the lipid, and dispensing these mixtures into the 384-well plates. Detailed standard operating procedures for each method, along with relevant instrument settings and data analysis spreadsheets, are available in the supplementary materials.

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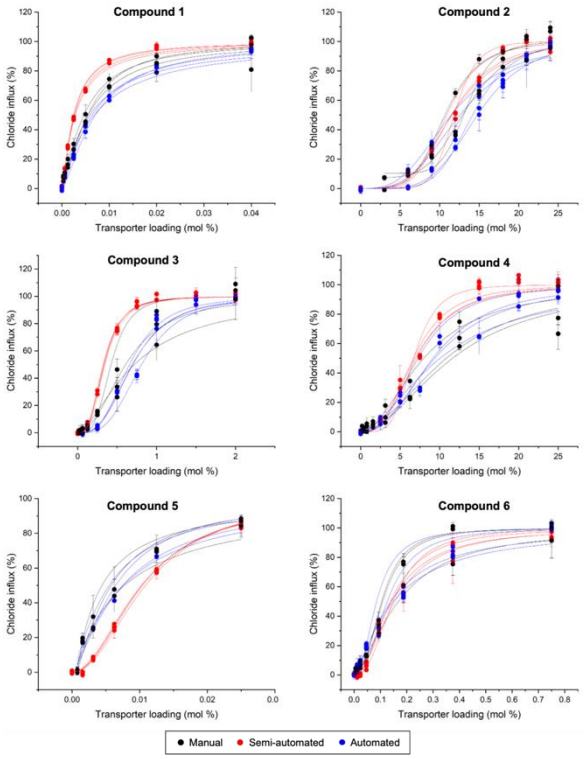


Figure 3: Compiled Hill plots for compounds **1**–**6.** Black curves represent results from the manual HT lucigenin assay where 3 repeated Hill plots were performed in 3 different batches of vesicles. Error bars represent standard deviation. (b) Red curves represent results from the semi-automated method where 4 repeated Hill plots were obtained from lipid and transporter solutions that were prepared by hand, and automatically dispensed into the plate. (c) Blue curves represent results from the automated method where 4 repeated Hill plots were obtained, and all pipetting was performed by the Opentrons.

The manual 96-well plate method exhibited the greatest variation in results, likely due to inconsistencies in manual pipetting and batch-to-batch variability among the vesicle samples. The standard deviations of the three EC_{50} values derived from replicate Hill plots for each compound ranged from 8% to 37% of the mean value (**Figure 4a**). While repeat Hill plots are infrequently reported in the literature, these error margins align with the variability observed in a large compound library assessed using ion-selective electrode assays, which reported errors between 9% and 38%.[22] These findings underscore the inherent variability in EC_{50} measurements across experiments, highlighting the necessity for efficient protocols to facilitate repeat experiments.

Compounds **2** and **5** emerged as the most challenging ionophores to study within our series. For compound **5**, the highest loadings (>0.02 mol% w.r.t. lipid) resulted in unexpectedly low transport activity (**Figure S30**), necessitating the exclusion of these data points for fitting to the Hill equation. Our EC_{50} values for compound **5** were the least reproducible, with errors reaching up to 38%, while the largest discrepancy between literature and high-throughput (HT) EC_{50} values was observed for compound **2**. We attribute these issues to difficulties in delivering these compounds to the vesicle bilayer. The low transport activity of compound **2** required testing at very high concentrations, which could have resulted in aggregation. Additionally, squaramides like compound **5** are poorly soluble in solvents other than DMSO due to the potential for intermolecular hydrogen bonding. [28–30] Previous studies have indicated that more dilute stock solutions of compound **5** in DMSO exhibit higher transport activity, suggesting a challenge in delivering this compound effectively.[28] While co-delivery with larger volumes of DMSO could enhance delivery into the bilayer, this approach is incompatible with our HT assay, which requires a constant DMSO volume to maintain consistent fluorescence quenching. Another significant distinction between our assay and the ion-selective electrode assay is the absence of stirring while adding the transporter solution. We hypothesise that the lack of stirring may exacerbate

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delivery issues. Further investigation of deliverability is critical for the potential biological applications, as some poorly deliverable ion transporters that demonstrate high activity in vesicle studies may show poor activity in cells.[14]

Overall, we have developed analogues of this assay with varying levels of automation. **Figure 4** illustrates the results from the manual, semi-automated, and fully automated HT methods, showing good agreement with literature values. The correlations between HT and literature EC_{50} values are visually represented, with R^2 values of 0.99, 0.98, and 0.89 for the manual, semi-automated, and automated methods, respectively. These strong correlations indicate that the HT assay produces accurate EC_{50} data comparable to that obtained through traditional methods.

A)

Compound	EC_{50} (published)	EC_{50} (manual)	EC_{50} (semi-automated)	EC_{50} (automated)
1	0.0048	0.0057 ± 0.00050	0.0029 ± 0.00017	0.0066 ± 0.00052
2	4.96-6.36	12.5 ± 1.5	11.8 ± 0.33	13.4 ± 1.3
3	9.83-11.05	9.3 ± 1.7	7.53 ± 0.15	9.55 ± 1.7
4	0.399-0.483	0.59 ± 0.15	0.34 ± 0.0056	0.70 ± 0.075
5	0.01	0.018 ± 0.0067	0.0098 ± 0.00031	0.0067 ± 0.00099
6	0.16	0.14 ± 0.039	0.15 ± 0.19	0.154 ± 0.00087

B)

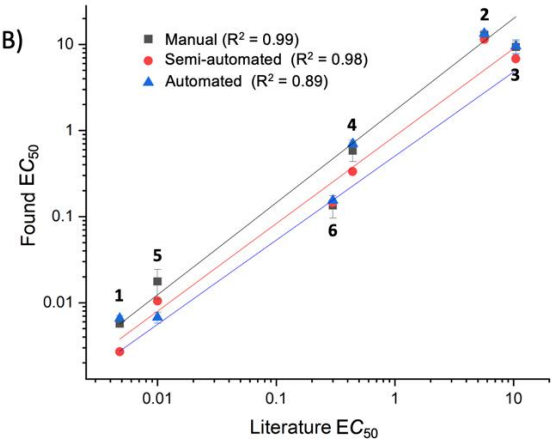


Figure 4: (a) A summary of the EC_{50} values determined via our high-throughput assay and (b) a visual representation of the HT EC_{50} values plotted against the literature EC_{50} values. Labels represent the compound tested, and is placed above or below the relevant data points. Error bars show the standard deviation of 3 repeats for the manual method, and the standard deviation of 4 experimental repeats for the semi-automated and automated methods. The R^2 values are shown in the legend.

Our methods demonstrate significant potential to save time and enhance data generation within anion transport workflows. Using the manual 96-well method, we were able to prepare up to six plates in a single day, yielding a total of 18 Hill plots. In contrast, conventional procedures for generating Hill plots typically allow for only one plot to be collected per day. The use of a plate reader streamlines the process, enabling the addition of sodium chloride followed by fluorescence measurement to be completed in approximately 10 minutes per 96-well plate.

The most time-consuming and labour-intensive aspect of this assay was the pipetting, prompting us to optimise this further. Our fully automated method allowed us to collect 24 Hill plots in a single day. Each 384-well plate required approximately two hours for preparation in a fully automated process that does not require researcher intervention after the method started, followed by 30 minutes for assessment on the plate reader to generate 12 Hill plots. Future improvements may be achievable by incorporating a liquid-handling robot equipped with multi-channel pipettes. This would facilitate the batch-wise addition of pre-prepared DMSO solutions to the vesicle-containing plates. This would not only expedite the addition of transporter solutions but also increase the number of plates prepared daily while minimising lipid degradation, given that the vesicles should be used within 2-3 hours. However, we did not explore this on the Opentrons OT-2 due to concerns about instrumental error for low-volume transfers, as the reported accuracy of the 8-channel GEN2 pipettes at 1 µL is ±20%. Maintaining the DMSO

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concentration at a precise 1% v/v is crucial, as any variation could interfere with results due to DMSO's quenching properties.

Conclusion

In conclusion, we have developed high-throughput methodologies for quantifying chloride/nitrate antiport activity using a lucigenin assay. Validation with six previously characterised anionophores demonstrated that the EC_{50} values obtained were consistent with literature reports. Notably, these methods significantly reduce the time required compared to traditional approaches, and the integration of the Opentrons robotic liquid handling system enhances efficiency while minimising user error, making our protocols accessible to researchers with varying levels of experience.

To our knowledge, this represents the first high-throughput approach for conducting a Hill plot analysis of anion transport. Our assay provides a foundation for further advancements in high-throughput methodologies. Given the current intense research interest in ion transporter design, we hope that our methods will accelerate the pace of discovery and translation in this area. The enhanced capacity for data generation will allow for the exploration of larger anionophore libraries. This broader exploration of chemical space will contribute to a deeper understanding of structure-activity relationships and aid in the identification of new design principles. By making these high-throughput methods accessible to the research community, we aim to alleviate the experimental bottleneck in ionophore screening, ultimately advancing research and facilitating the discovery of novel biologically active transporters.

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